

Evaluating Time-Series Foundation Models

Scientific Executive Summary

21 September 2026; results computed 17–21 September 2026

Main findings

- **Foundation models:** Chronos-2 achieves scaled MASE 0.6855 and wins 74 of 98 matched TIME tasks, ahead of TS-ICL and Chronos-Bolt. It also has the lowest measured total forecasting time (439.1 seconds).
- **Channel representation:** native multivariate Chronos-2 is 0.8% better in aggregate than independent univariate forecasting. Recasting other target histories as past covariates gives essentially the same accuracy as native input, with 5.68 times its measured forecasting time.
- **Dataset features:** greater temporal heterogeneity accompanies higher scaled MASE for all three foundation models, with Spearman rank correlations of 0.42–0.50 across the 50 configurations.
- **Context length:** Chronos-2 at 4,096 input steps has sMASE 0.6824, 0.45% lower than at 8,192, with 40.7% less recorded forecasting time. Halving the maximum context costs 1.34% sMASE for Chronos-Bolt and 0.21% for TS-ICL, while reducing time by 38.9% and 59.5%, respectively.
- **Instance normalization:** finite-value, per-window z-scoring followed by inversion changes no model’s aggregate sMASE materially; all three agree with unchanged input to at least five decimal places.

1 Datasets and test design

The evaluated TIME subset contains **39 distinct datasets**, represented by **50 dataset–sampling–frequency configurations**. A dataset can appear at several frequencies. Each configuration has one short-horizon task; 24 also have medium and long horizons, giving **98 forecasting tasks** in total ($50 + 24 + 24$).

- **Size.** Configurations contain 1–989 series, with 1–60 measured channels per series. Their mean full-series lengths range from 114 dates (Housing_Inventory/M) to approximately 68,605 dates (Coastal_T_S/5T). These are numbers of observations, not equal calendar durations.
- **What is a series versus a channel?** One series is a time-indexed table: rows are dates and columns are measured channels (variates). A dataset may contain several such tables, rather than one large shared spatial axis. SG_Weather/D, for example, contains 6 separate tables, each with 4 channels. 16 of the 50 configurations contain both several tables and several channels per table. A forecast window is taken from *one* table.
- **Splits.** The official configuration reserves a final test segment of 20–4,320 dates and a preceding validation segment of 8–4,320 dates; the remaining prefix is the training segment. These lengths vary by configuration, rather than following one percentage split. In this frozen model evaluation, all observations before a forecast origin are available as history, including the preceding validation segment.
- **Forecasts.** The test segment is partitioned into complete, nonoverlapping windows. Horizons range from 3 to 864 dates, with 1–186 test windows per series and task. Earlier test observations become available at later origins; future targets do not.

2 Forecasting task and evaluation

Let x be the input window from one series. It can be univariate or multivariate: it contains one or several measured channels, over the most recent dates available at the forecast origin. It uses up to the model’s maximum context length—for example, **8,192 dates for Chronos-2**— and fewer if the available history is shorter.

Let y contain the next H dates of those channels, where H is the forecast horizon. A frozen pretrained forecaster f produces a median forecast $\hat{y} = f(x)$ and quantiles 0.1, 0.2, $\dots$, 0.9. Evaluation compares $\hat{y}$ with y . Metric scaling uses the full history available at the origin, not only the bounded model input x .

Mean absolute error (MAE). For one channel, let A be the set of future positions with finite observations. At position $h \in A$, y_h is the observed value and $\hat{y}_h$ the median prediction. Then

$$\text{MAE}(y, \hat{y}) = \frac{1}{|A|} \sum_{h \in A} |y_h - \hat{y}_h|,$$

where $|A|$ is the number of evaluated positions and the absolute value measures the magnitude of each prediction error.

1. Historical seasonal MAE: the scale of one forecast error. Let $z_1, \dots, z_T$ be the full history of this channel at the forecast origin, T its length, and p the frequency-specific seasonal lag. Define $J = \{j : p < j \leq T, z_j \text{ and } z_{j-p} \text{ are finite}\}$. Then

$$\text{MAE}_{\text{seasonal, history}} = \frac{1}{|J|} \sum_{j \in J} |z_j - z_{j-p}|, \quad \text{MASE} = \frac{\text{MAE}(y, \hat{y})}{\text{MAE}_{\text{seasonal, history}}}.$$

This seasonal scale is computed on *history*; it is not the baseline’s test error. Missing dates remain at their original positions.

2. Arithmetic task means: average the window–channel errors. One metric cell c is one channel in one forecast window of one series. For task k , let G_k be its shared set of eligible cells. For method m , write $\text{MASE}_m(c)$ for the cell score defined above. The task mean is

$$\overline{\text{MASE}}_m(k) = \frac{1}{|G_k|} \sum_{c \in G_k} \text{MASE}_m(c).$$

Every cell has equal weight, regardless of its series or channel. G_k requires a finite future observation, a positive finite historical scale, and finite Seasonal test error. Learned quantiles must be finite on this support. Seasonal Naive repeats the latest season, forward-filling history gaps and using the first observation for leading gaps.

3. Task scaling, then a geometric mean across tasks. The matched Seasonal task mean is $\text{MASE}_{\text{seasonal}}(k) = \overline{\text{MASE}}_{\text{seasonal}}(k)$. For the learned method m , define

$$\text{sMASE}_m(k) = \frac{\overline{\text{MASE}}_m(k)}{\text{MASE}_{\text{seasonal}}(k)}, \quad \text{sMASE}_m = \exp\left(\frac{1}{K} \sum_{k=1}^K \log \text{sMASE}_m(k)\right),$$

where K is the number of matched tasks (98 or 74 in these experiments). The task score is a *ratio of arithmetic means*, not an average of cell-wise model/Seasonal ratios. The reported sMASE gives every task equal weight; lower is better and Seasonal equals 1.

Configuration dispersion. Apply the same geometric mean within each configuration’s available horizons. The 25th–75th percentile interval across the 50 configuration scores contains their **middle 50%**; its width is the **interquartile range (IQR)**. It measures cross-configuration dispersion, not repeated-run uncertainty.

3 Foundation-model accuracy and forecasting time

Question. Which of the evaluated frozen models gives the lowest forecast error relative to Seasonal Naive, and at what measured forecasting cost?

Protocol.

- **50 configurations, 98 tasks:** 38 multivariate and 12 univariate configurations; 50 short, 24 medium, and 24 long horizon tasks. Prediction lengths range from 3 to 864 steps.
- **Models:** Chronos-2, Chronos-Bolt Base, and TS-ICL v1, with context caps of 8,192, 2,048, and 4,096 observations respectively. Forecast batches contain at most 16 input items. Chronos-2 groups variates as targets; the other two models forecast variates independently.

Model	sMASE	Middle 50% of configs	Configs better than Seasonal	Task wins	Seconds
Chronos-2	0.685489	0.612–0.774	48/50	74/98	439.1
TS-ICL	0.718394	0.673–0.808	47/50	20/98	2,672.5
Chronos-Bolt	0.759559	0.694–0.842	47/50	4/98	942.1
Seasonal Naive	1.000000	1.000–1.000	—	—	654.6

Table 1: The two counts have different units: “Configs better than Seasonal” is out of **50 configurations**, after aggregating their horizons; “Task wins” is out of **98 horizon tasks**, counting the lowest MASE among the three learned models. The middle-50% interval gives the 25th and 75th percentiles of configuration scores.

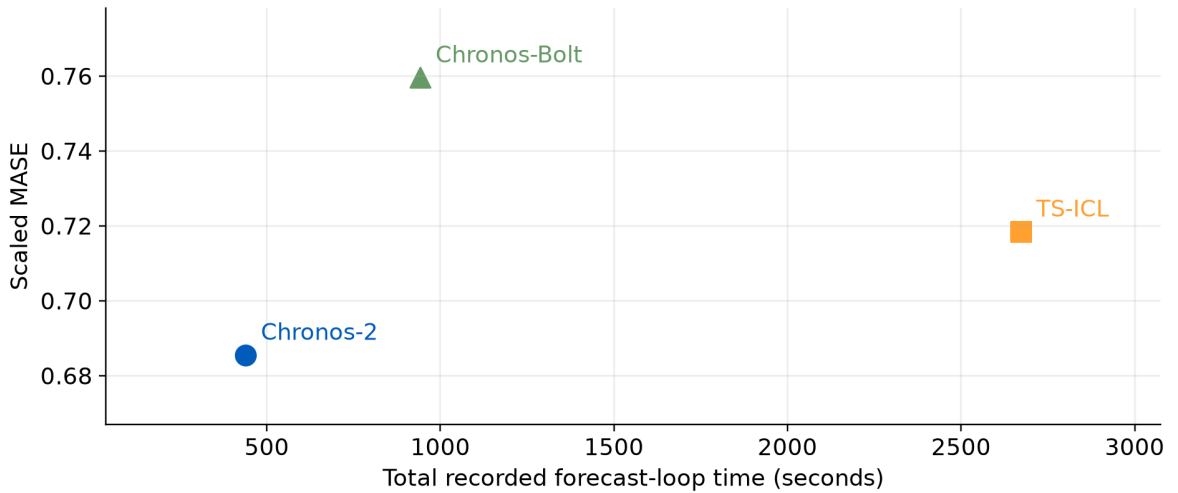


Figure 1: Accuracy and total recorded forecast-loop time on the 98 tasks. Lower and left is preferable. Time includes loop preparation but excludes model loading, metric computation, and saving.

Results.

- Chronos-2’s aggregate sMASE is **31.5% better than Seasonal**, 4.6% better than TS-ICL, and 9.8% better than Chronos-Bolt (lower error). It leads on 74/98 tasks and beats Seasonal on 48/50 configuration scores.
- TS-ICL beats Chronos-2 on 21 tasks, and Chronos-Bolt on 8. These pairwise counts differ from winning against *both* competitors. Their total forecasting times are 6.09 and 2.15 times Chronos-2’s.

Scope. All three models complete all tasks, with 111,071 eligible window–variate cells across items, out of 111,177 candidates. These are metric cells, not future time steps. Context lengths and wrappers differ; Seasonal uses CPU and learned models use accelerator inference. These times are not hardware-independent backbone latencies.

Within-task error dispersion

The task mean hides differences between forecast windows and channels. Using the same eligible cells G_k , define the **population variance** of task k ’s cell errors for method m :

$$\text{Var}_m(k) = \frac{1}{|G_k|} \sum_{c \in G_k} [\text{MASE}_m(c) - \overline{\text{MASE}}_m(k)]^2.$$

The denominator is $|G_k|$, not $|G_k| - 1$: these are descriptive statistics of all evaluated cells, not estimates from repeated experimental runs.

For comparison with Seasonal Naive, define its variance on the same cells as $\text{Var}_{\text{seasonal}}(k)$, using the formula above. The plot uses

$$\text{scaled mean} = \frac{\overline{\text{MASE}}_m(k)}{\text{MASE}_{\text{seasonal}}(k)}, \quad \text{relative variance} = \frac{\text{Var}_m(k)}{\text{Var}_{\text{seasonal}}(k)}.$$

The second quantity is a ratio of variances, not the variance of scaled MASE.

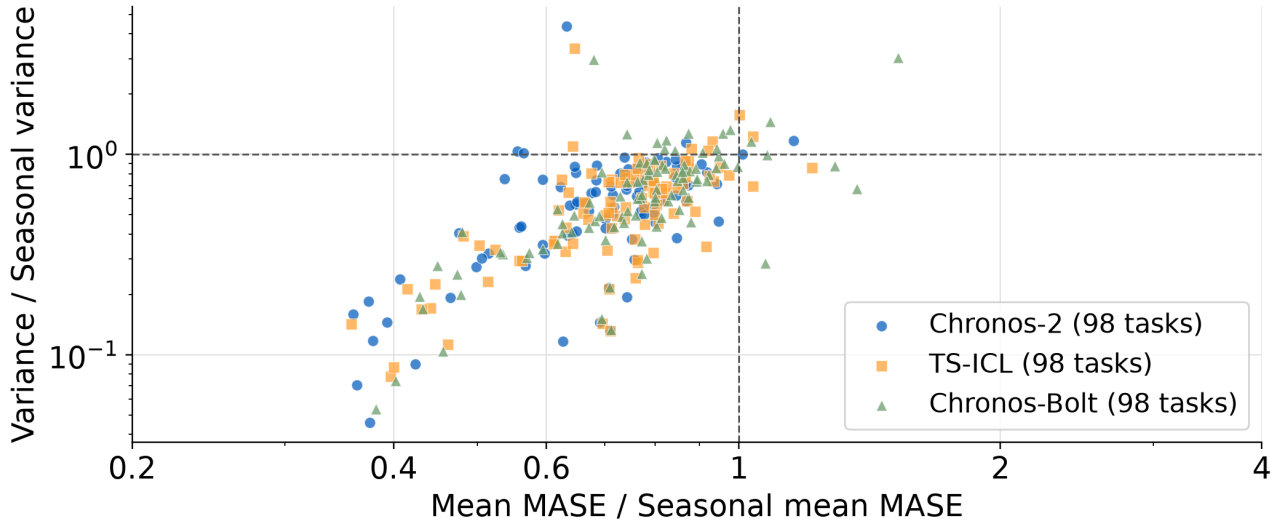


Figure 2: One point per selected model–task pair: **98 tasks per model, 294 points total**. The horizontal coordinate is the scaled task mean MASE; the vertical coordinate is the population MASE variance divided by matched Seasonal variance. Both axes are logarithmic and retain every point. Dashed lines mark Seasonal parity at 1: left means better mean error, and below means lower dispersion across windows and channels. Above 1 means greater dispersion than Seasonal, not an invalid score.

Observed dispersion. Median variance ratios across the 98 tasks are **0.590 for Chronos-2, 0.580 for TS-ICL, and 0.629 for Chronos-Bolt**. Variance is lower than Seasonal on 93, 91, and 84 tasks, respectively; both mean error and variance improve on 92, 89, and 80 tasks. Chronos-2’s lowest aggregate mean error therefore does not imply the lowest median relative variance. The model clouds overlap, and some tasks have greater dispersion than Seasonal. Lower variance alone does not guarantee better forecasts: an error can be consistently large.

4 Chronos-2 channel representation

Question. How does representing the same target histories jointly, independently, or as past covariates affect accuracy and forecasting time?

For a multivariate window x , write x_v for the history of variate v and x_{-v} for the histories of all other variates. The three modes are:

- **Native multivariate:** supply x as one grouped target input, and forecast all variates together.
- **Independent univariate:** supply x_v alone, separately for each variate.
- **Past-target covariates:** supply x_v as the target and x_{-v} as past-only covariates, separately for each variate. This carries the same historical information as native input through another interface; no future values are supplied.

Protocol. All modes use frozen Chronos-2, the 8,192-step context cap, at-most-16-item batches, identical quantiles, test windows, and Seasonal support. The matched subset has **38 multivariate configurations and 74 tasks**. Its native score is not the 98-task foundation score.

Representation	sMASE	Change vs native	Seconds	Time ratio
Native multivariate	0.690045	reference	340.0	1.00
Independent univariate	0.695590	+0.80%	350.2	1.03
Past-target covariates	0.690045	equivalent	1,930.8	5.68

Table 2: Matched 74-task sMASE and forecast-loop time. Positive score change is worse. Time ratios divide total seconds by native total seconds.

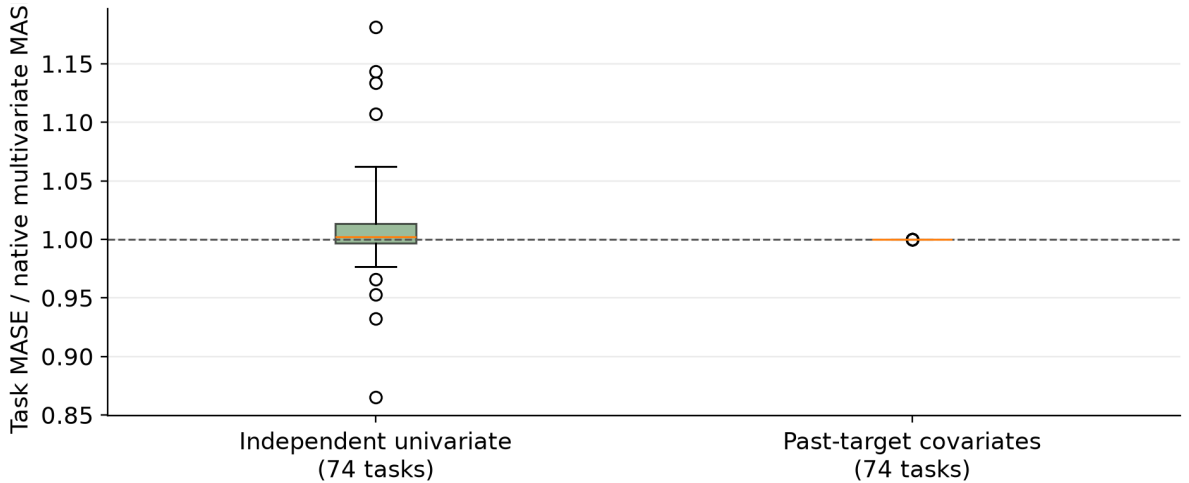


Figure 3: Task MASE divided by matched native MASE, 74 ratios per mode. Below 1 favors the alternative mode. Boxes show the middle 50% and median; whiskers span the 5th–95th percentiles, with remaining tasks shown as dots. These distributions describe differences across tasks.

Results and interpretation.

- Native input wins 46 tasks and independent input wins 28. The geometric mean independent/native ratio is 1.008; its middle-50% task interval is 0.997–1.014. Native input’s **0.8% aggregate advantage is small and varies across tasks**.
- Past-target covariates reproduce native MASE: the largest absolute relative difference is below 5.75×10^{-7} . They take **5.68 times the forecasting time**, without an accuracy gain in this comparison.

All modes complete all 74 tasks, using 91,319 eligible metric cells out of 91,425 candidates.

5 Dataset features associated with relative forecast error

Question. Which measured dataset characteristics accompany higher or lower sMASE across the evaluated configurations?

Protocol. Each model’s configuration sMASE is paired with full-series feature summaries, giving 50 configurations per model. Features are computed per source variate and averaged over finite values within each configuration. Spectral/decomposition preprocessing standardizes and interpolates observations. STL (seasonal–trend decomposition by locally estimated smoothing) uses the strongest detected candidate period, with a nonseasonal smoothing fallback.

Reported feature definitions.

- **Temporal heterogeneity:** split a series into four chronological blocks and average three components, each clipped to $[0, 1]$:
 - *Location:* variance of block means divided by whole-series variance.
 - *Scale:* $1 - \exp(-\sigma^2)$, where σ^2 is the variance of log block standard deviations relative to whole-series standard deviation, regularized by 10^{-12} .
 - *Spectral:* mean base-2 Jensen–Shannon divergence between each block’s normalized power spectrum and the mean spectrum. This divergence is mixture entropy minus mean individual entropy; spectra use 32 frequency bins and a constant-signal bin. Series shorter than 16 steps receive zero temporal heterogeneity.
- **Spatial heterogeneity:** compare all source channels across all series in a configuration, using each channel’s full history, rather than chronological blocks. “Spatial” here means differences between channels, not geographic distance. Average three components clipped to $[0, 1]$:
 - *Location:* variance of channel means divided by the sum of that variance and the mean within-channel variance.
 - *Scale:* $1 - \exp(-\sigma^2)$, now with σ^2 the variance of log channel standard deviations relative to their median standard deviation, regularized by 10^{-12} .
 - *Spectral:* the same Jensen–Shannon comparison, between full-channel spectra and their mean spectrum. A configuration with fewer than two source channels has zero spatial heterogeneity.

Other feature names in the table link to their implementation or estimator.

Feature results and interpretation

Feature	Chronos-2	TS-ICL	Chronos-Bolt	Configs
Temporal heterogeneity	+0.497	+0.486	+0.418	50
Spatial heterogeneity	−0.036	−0.029	+0.035	50
Trend Hurst	+0.438	+0.436	+0.360	50
Temporal scale heterogeneity	+0.442	+0.418	+0.332	50
Temporal location heterogeneity	+0.409	+0.397	+0.369	50
Second detected period	−0.429	−0.368	−0.332	39
Seasonal strength	−0.352	−0.254	−0.132	50
Seasonal cycle correlation	−0.280	−0.202	−0.168	50
Stationarity	−0.128	−0.280	−0.242	50
Spectral entropy	−0.192	−0.277	−0.356	50
Spikiness	−0.166	−0.142	−0.272	50

Table 3: Spearman rank correlations with configuration sMASE: Pearson correlation of ranks, assigning average ranks to ties. Bold entries are each model’s top three positive or top three negative correlations among all extracted features; rows are their union plus spatial heterogeneity. Positive means larger features accompany worse relative error. Counts use pairwise finite configurations.

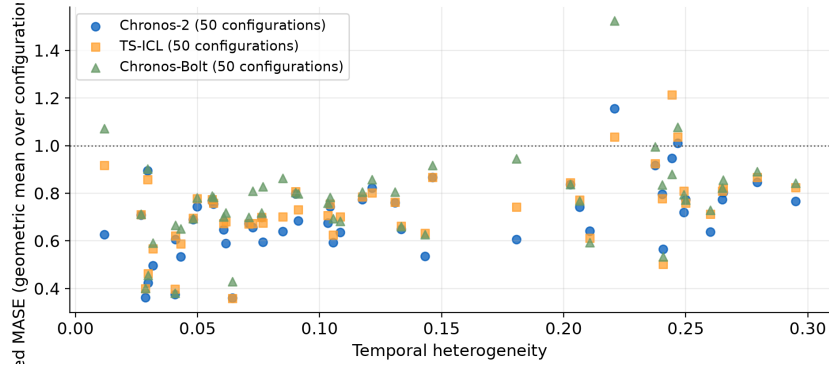


Figure 4: Temporal heterogeneity versus configuration sMASE, replotted from the completed feature-analysis data: 50 points per model in one panel. The dotted line marks Seasonal parity.

Interpretation. Greater temporal heterogeneity and trend Hurst accompany worse relative forecast error for all three models. Spatial heterogeneity has near-zero correlations (−0.036 to +0.035), unlike temporal heterogeneity. Negative leaders vary by model: second detected period, seasonal strength/correlation, stationarity, spectral entropy, and spikiness. These cross-dataset associations do not isolate causal effects. The displayed features were selected from the extracted feature set and use full series, including test observations, so they describe the obtained errors rather than an evaluated pre-forecast model-selection rule.

Evidence scope. Selected task records and metric summaries were checked; raw arrays were not. The IQRs and correlations describe this benchmark, not repeated-run variability.

6 Maximum input context length

Question. How much history should each frozen forecaster receive? Longer inputs may improve accuracy but increase inference work. The tested factor is the maximum retained input length, not the forecast horizon.

Protocol. For each of Chronos-2, Chronos-Bolt, and TS-ICL, the model-specific maximum is successively halved four times. Each of the 15 model–context settings completes the same 98 TIME tasks on the same 111,071 eligible window–channel cells. The input at each forecast origin is the most recent available history up to the selected cap; shorter histories remain shorter. Model weights, output quantiles, official test windows, Seasonal support, and sMASE aggregation are unchanged. There is one selected run per setting, not repeated-seed uncertainty.

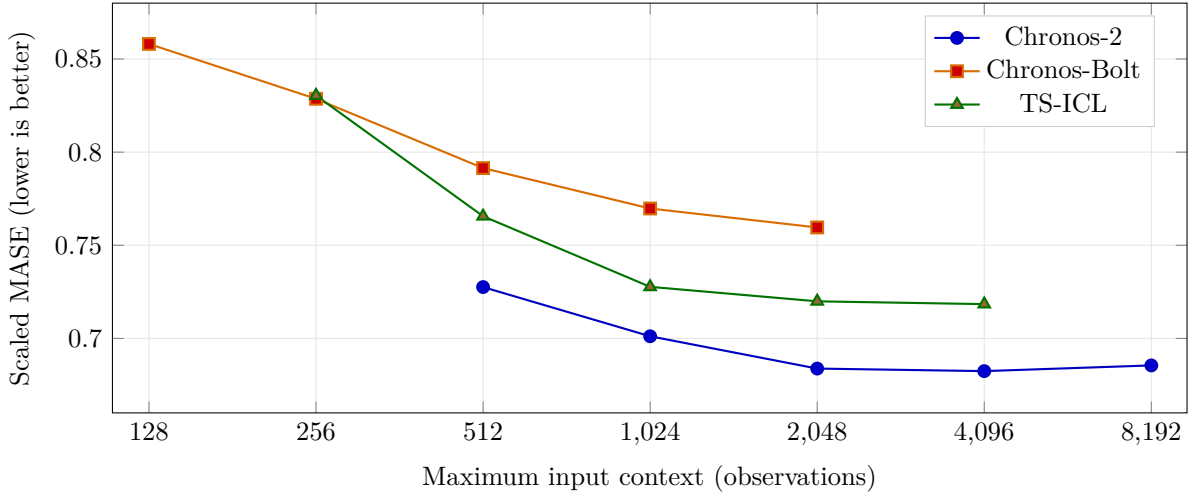


Figure 5: Aggregate sMASE versus maximum input context for each frozen model. Each point is the geometric mean of 98 Seasonal-scaled task MASE ratios on the shared evaluation grid; lines connect only the five tested settings per model and do not establish behavior between them. Context is on a base-2 logarithmic axis; lower sMASE is better.

Results. Chronos-2 at 4,096 steps improves aggregate sMASE by 0.45% and reduces recorded time by 40.7% relative to 8,192. At 2,048 steps, its sMASE is still 0.25% lower, with 63.9% less time. The paired 4,096-versus-8,192 comparison improves 49 tasks, ties exactly on 28, and worsens 21; its median task change is only -0.01% . Thus the aggregate advantage is modest and not uniform. Chronos-Bolt and TS-ICL attain their lowest sMASE at the longest tested context. Halving that context worsens their aggregate scores by 1.34% and 0.21%, respectively, while reducing recorded time by 38.9% and 59.5%. Their shortest settings worsen sMASE by 12.97% and 15.59% relative to their maxima.

Limit. Each timing is a single-launch forecast-loop total, not a controlled repeated latency estimate or an end-to-end job duration. The comparison supports a model-specific accuracy–cost tradeoff on these tasks, not a universal optimal context length.

7 Per-instance input z-score normalization

Question. Does centering and scaling each retained input window improve frozen-model accuracy when forecasts are returned to the target’s original units?

Protocol. Compare unchanged input with z-scored input for each of the three models at its maximum context (8,192, 2,048, and 4,096 steps, respectively). For each window and variate, compute the finite-value mean and population standard deviation along time after context truncation; constant finite inputs use unit scale. Missing positions remain missing. Every predicted quantile is inverse-transformed before scoring. Each of the six model–mode settings completes the same 98 tasks and 111,071 eligible cells. The unchanged-input arms exactly reproduce the matching task metric summaries from the maximum-context evaluation.

Model	sMASE ↓		Forecast-loop seconds ↓	
	Unchanged	Z-score	Unchanged	Z-score
Chronos-2	0.685488615	0.685488629	434.6	431.7
TS-ICL	0.718393983	0.718393972	2,681.1	2,686.8
Chronos-Bolt	0.759559036	0.759558465	945.6	1,001.6

Table 4: Matched 98-task geometric sMASE per model and mode. Seconds are single-launch forecast-loop totals, not a controlled speed comparison.

Result and interpretation. Z-scoring provides no material aggregate accuracy gain for these frozen models on this benchmark. Every model’s two scores agree to at least five decimal places, and the largest relative difference on any paired task is 0.0061%. This near-invariance concerns the tested input/output transformation; it does not imply all normalization methods are interchangeable. An entirely missing retained window has no finite mean or scale and remains non-finite rather than receiving an invented signal; the completed selected tasks provide no evidence of such a window reaching a model. Timing differences from one launch do not establish a normalization speed effect.